LUNAR

DESIGN & ENGINEERING PORTFOLIO

2024 F1 in Schools World Finals | Saudi Arabia

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DESIGN OVERVIEW & CAR BEHAVIOUR

OVERARCHING OBJECTIVES

1. Compliance	The car needs to be 100% compliant with all regulations.
2. Performance	The car needs to be as fast as possible.
3. Durability	The car needs to withstand all races without breakage.

ENGINEERING GOALS

Goal 1: Weight	<50.08g
Goal 2: Low drag	<0.28N
Goal 3: Stability	<0.11s varience between race times

'3 Double D' Design Philosophy

Engineering is an iterative Design Thinking Process and continuous cycle of improvement. With this in mind we have developed our own '3 Double D' design philosophy with 3 overarching engineering objectives.

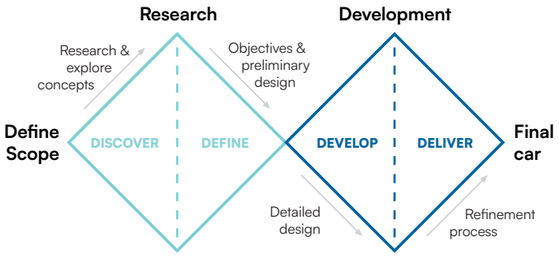
›› Data Driven results

All development decisions much be backed with quantifiable statistical improvement.

›› Deliberate Design

Each design is carried out with clear purpose linked to achieving our overarching objectives through first principles.

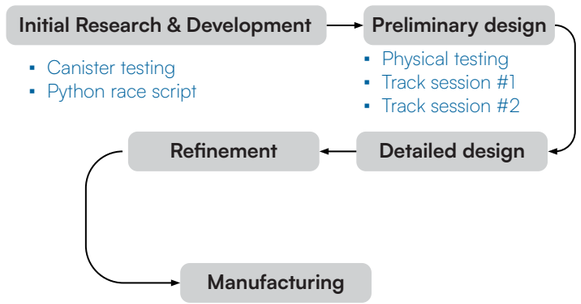
›› Double Diamond R&D cycle:



2.2: The double diamond R&D model for our car
Credit: Lunar & UK design council

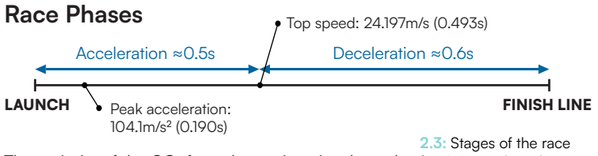
We believe having a design philosophy allows us to integrate human centred, ethical and forward-thinking approach into our race car.

Our Process and Related Testing



CAR BEHAVIOUR RESEARCH

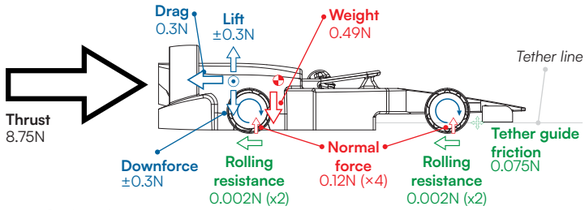
Race Phases



The majority of the CO_2 from the canister is released from the puncture at the launch stage which is mostly **uncontrollable** within the regulations.

2.3: Stages of the race (statistics gathered from National Finals car using our mathematical model)
Credit: Lunar

Free Body Diagram

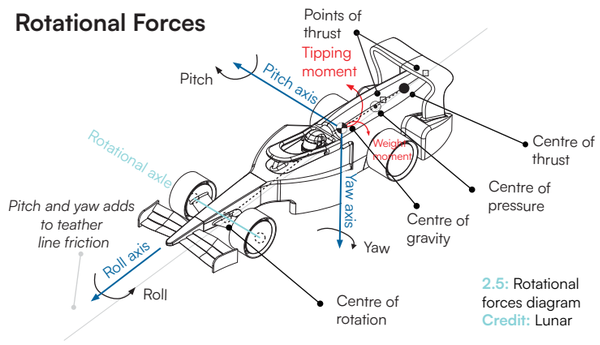


= Canister thrust
= Aerodynamics/centre of pressure
= Mass/centre of gravity
= Friction

*Approximate figures

2.4: Free body diagram
Credit: Lunar

Rotational Forces



2.5: Rotational forces diagram
Credit: Lunar

Variables Impacting Car Performance

Controllable	Uncontrollable
▪ Mass (F=ma) ▪ Aerodynamics (drag & stability) ▪ Cross-sectional area ▪ Drag coefficient ▪ COP placement, vertical & horizontal aerodynamic stability ▪ Stability ▪ Tipping moment (pitch) ▪ COG placement ▪ Center of thrust (canister tilt) ▪ Yaw ▪ Wheel system ▪ Rotational inertia ▪ Wheel friction ▪ Wheel stability ▪ Bearing friction ▪ Wheel strength ▪ Tether guide ▪ Friction ▪ Stability	▪ Force produced by canister ▪ Volume of CO_2 ▪ Temperature of canister ▪ Puncture size ▪ Track surface (frictional difference) ▪ Aerodynamic instability ▪ Room conditions ▪ Temperature ▪ Humidity ▪ Canister puncture position ▪ Raw canister weight (without CO_2) ▪ Breakages during the race

Impact on race time

Sub-systems
- Wheel system
- Tether guideline system

CAR BEHAVIOUR MODELLING

**Pre-design testing & research*

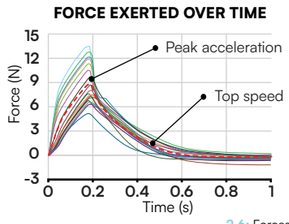
Aim: To determine the extent of the impact of various variables on the performance of the cars.

Launch Behaviour

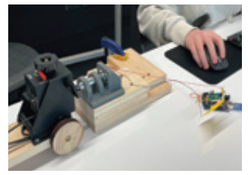
›› Canister force testing **Physical testing*

Aim: To understand the total amount of forwards force on the car and acquire data for python race simulation.
Method: To measure the thrust on the car, we developed a custom load cell jig connected to an audrino board (see figure 2.7), measuring the force exerted by the canister 89 times per second. We tested 18 canisters (weighed between 28.5 and 29.5 grams) and graph the results below.

Outcomes: Using the mean forces at each time sector, the average demonstrates a sharp rise to a peak of 8.75N around 0.2 seconds in, followed by a gradual descent to 0N 0.5 seconds in.



2.6: Forces exerted by 18 canisters (left).
Credit: Lunar



2.7: Canister testing load cell jig (top)
Credit: Lunar

›› Canister varience

We discovered that the force produced by the canisters varied hugely (between 6.27 and 13.4 N) depending on the **CO_2 volume, puncture hole size and room temperature.** Factors such as the puncture position also has the potential to cause yaw. To offset this difference in our maths model, we implemented a piecewise function modelling canister behaviour (see below).

Python Race Simulation

The python simulation we created allowed us to mathematically calculate the theoretical performance of a car given key statics and variables.

The model we created:
- Uses preset formulas to caculate and subtract non-propelling forces from the canister force
- Divides the mass and rotational inertia to find acceleration
- Then calculates the velocity and distance over time to output an estimated race time

The model also features a piecewise function breaking the phases of the race down to intervals of 0.00, 0.05, 0.19, 0.23, 0.525 seconds to adjust the canister force based on physical testing, accounting for changes in canister mass as air escapes, which reduces the car's weight during travel.

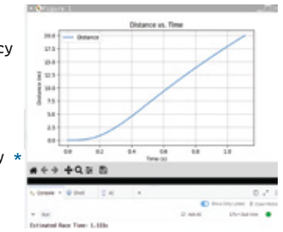
Other variables dependant on race conditions include rolling resistance considering wheel deformation and radius along with an adjustable yaw value. As the race progresses, yaw is dampened by a factor of 0.6 to mimic to the tether pulling the car toward the center of the track.

Users input various parameters, including:
- Car mass
- Drag force
- Lift force
- Wheel friction coefficient
- Yaw
- Tether guide friction coefficient
- Wheel radius
- Rotational inertia
- Length of the wheel in contact with the track

This outputs information in graphs (see below) to tell us an accurate estimation of our car's theoretical results car that we could later apply to our prototypes.

```
# Update yaw angle
yaw_angle = yaw_amplitude *
math.sin(2 * np.pi * yaw_frequency
* time) * math.exp(- time /
damping_factor)

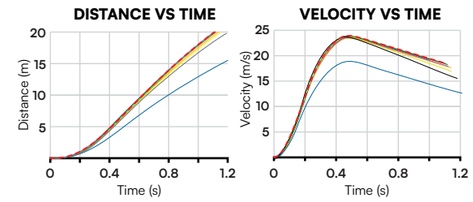
#Calculate Rolling Resistance
rolling_resistance_coefficient =
((mass + canister_mass) * gravity *
wheel_contact_patch_size) /
math.sqrt(wheel_radius**2 -
wheel_contact_patch_size**2)
```



2.8: Python Race Simulation script and output
Credit: Lunar

›› Simulation Results

By amplifying each variable by 50% and seeing the changes in overall times, we are able to evaluate the impact that controllable variable has on race times. This helps us narrow down on specific areas to consider when developing our current car.

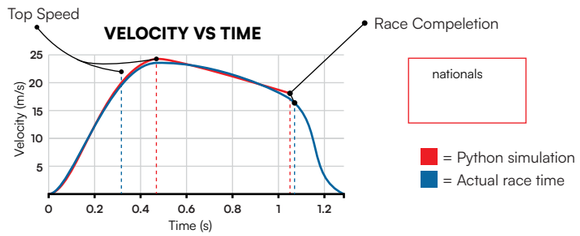


= Standard national finals car
= Wheel radius (50% increase)
= Wheel friction (50% increase)
= Tether guide friction (50% increase)
= Rotational inertia (50% increase)
= Mass (50% increase)
= Lift (50% increase)
= Drag (50% increase)

Limitations: This model is not able accurately reflect the impact of rotational forces on the overall performance of the car as it neglects its impact on canister efficiency. It had to be supplemented with our COG and COP physical testing.

Accelerometer Testing **Track session #1*

To validate our race model, we attached an accelerometer to the underside of our National's car, it was raced down the track three times while the math model's canister peicewise function was adjusted to match the performace of the canister on that day. The results from the testing can be roughly described by the graph below.



3.3: Car variables with highest impacts
Credit: Lunar

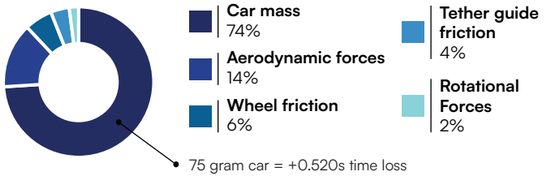
Total mathematical time: 1.125s
Actual average race time : 1.136s

PRE-DESIGN TESTING & RESEARCH

Summary of Findings

By studying the behaviour of the car in the race, we are able to determine the order of importance of all controllable variables and components that impact car performance. This helps us to avoid issues such as “sandboxing” (where systems are isolated) and allows us to better priortise the research and development of certain elements of the car over others.

IMPACT OF CONTROLLABLE VARIABLES



3.2: Estimated percentage impact of controllable variables on total performance
Credit: Lunar

Performance Objective Sub-goals

Findings from our pre-design testing and reasearch along with past experience allows us to define and create sub-goals for the car development process.

Goal #1: Car Mass (74% performance)*

Increasing the car’s mass directly decreases its acceleration and speed as shown by Newton’s 2nd law (F=ma), this is proven with the math model as it had the biggest impact on car performance.

Goal: To have a car at the minium weight limit
KPIs: To be at 50.08 grams — 0.08 gram tolerance

Goal #2: Aerodynamics (14% performance)*

Aerodynamic efficiency: The only two ways of reducing the drag force on a car is through reducing its frontal area and reducing its drag coefficient, shown by the following formula:

Drag force (N) — Velocity **Uncontrollable** — Cross-sectional area **Controllable** — Fluid density **Uncontrollable** — Drag coefficient **Controllable**

$$F_D = \frac{1}{2}\rho v^2 C_D A$$

Aerodynamics are also able to affect the stability of the car with aerodynamic elements being able to be used to minimise yaw or generate lift.

Goal: Minimise frontal area and drag coefficient
KPIs: less than XCd drag coefficient
achieve less than Xcm² frontal area

Goal #3: Stability (12% performance)*

The directional stability in the form of pitch, roll and yaw affects the car in unpredictables ways, decreasing the overall efficiency of the thrust force produced by the canister effects the tipping moment whilst creating inconsistent race results.

Goal: Minimise rotational forces through creating a stable car and create a stable and low friction wheel system
KPIs: COG placement in line with center of thrust create 0.05N lift at 20m/s to offset tipping moment, minimise yaw through longer wheelbase, priortise consistency over single lap speed in track testing

**Values adjusted to disregard sub-systems, stability also affects the consistency of car performance*

PRE-DESIGN RESEARCH

Weight Distribution & Stability

The centre of gravity (COG) is an imaginary point of a solid where the total weight is thought to be concentrated.

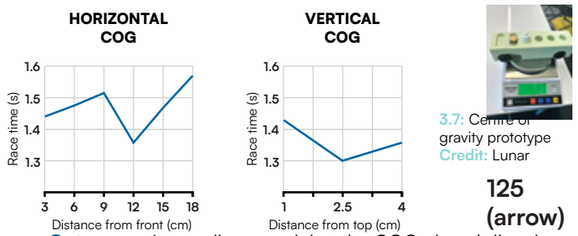
Centre of gravity testing

**Track Session #1*

Aim: To understand the impact of the distribution of mass on the car’s performance.

Hypothesis: Our theory was that additional mass concentrated slightly above the canister’s line of thrust can help offset the car’s tipping moment.

Method: A test car that weighted exactly 100 grams was designed with and additional 25 grams of weight added in the form of four two dollar coins on various positions of the car. This design allowed for 14 different configerations of different horizontal and vertical placements. The test car was sent down the track a total of 42 times ensuring that each position was track tested three times for accurate results.



3.7: Centre of gravity prototype
Credit: Lunar

125 (arrow)

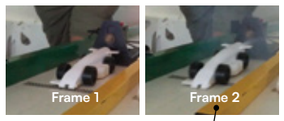
Outcomes: It was discovered that the COG placed directly in front of the canister’s line of thrust as far back as possible produced quicker and more consistent results.

Relevance to car: We tried to design a car that is as light as possible to save weight for the potential to create a heavier 3D printed halo, which is loacted precicely where we placed the additional weight for better COG.

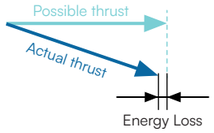
Tipping Moment

The tipping moment is a observed to be a direct result of the misalignment between the center of thrust causing the car to pitch forward wasting massive amouts of canister thrust, whilst causing the car to become unbalanced.

Relevance to car: We aim to design a rear heavy car with a higher COG as close to the center of thrust as possible. We also experimented with creating a lift generating front wing to offset this effect.



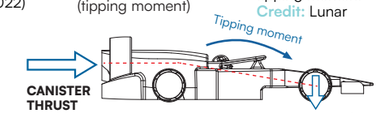
3.8: Tipping moment track test
Credit: Lunar (2022)



3.9: Energy lost from tipping moment
Credit: Lunar

Car touching track (tipping moment)

3.10: Tipping moment forces diagram
Credit: Lunar



Yaw

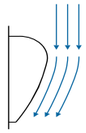
Yaw is a very common occurrence during races, and is largely caused by an uncentered canister puncture position. A car with a longer wheelbase is more resistant to yaw.

Aerodynamic Efficiency

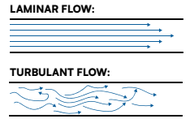
Aside from decreasing the frontal aera of the car, we need to reduce the car’s drag coefficient, to do this, we need to have an understanding of what causes drag.

Skin friction drag

Relevance to car: Skin friction drag causes air to stick to the surface of the car due to friction with air particles, curved surfaces encourage laminar flow through the ‘coanda effect’.



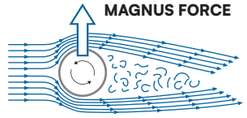
To minimise skin friction drag:
- Smooth surfaces
- Maintain laminar flow
- Low total surface area



Pressure drag

Relevance to car: Flow separations caused by the car create high and low pressure zones know as “adverse pressure gradients” creating wake. These pressure zones cause far more drag than skin friction drag, and hence it is important for us minimise pressure zones.

The car’s wheels also create massive amounts of wake through the “magnus effect” when spinning in high speeds, it is integral to direct airflow away from wheels and manage wheel wake.



3.4: Wake left by spinning wheel
Credit: Lunar

To minimise pressure drag
- Create a streamline shape e.g. teardrop
- Reduce airflow contact with wheel
- Create smooth transitions between components to decrease flow seperation

Aerodynamic Stability

Centre of pressure testing

**Track Session #1*

Aim: To determine the effect of COP on car performance.

Hypothesis: It was hypothesised that the COP has minimal impact on car performance but should be placed as far forward as possible to generate lift.

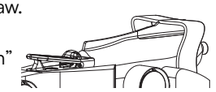
Method: An aerodynamically symetric test car (43.7 grams) was created with 21 pins drilled to the top. Allowing a fin to be plugged into any position on the car. The fin creates high-pressure zones, thus helping determine the location of the CoP. We track tested each position three times with the results described below.



PIN POSITION	AVERAGE
1-3 (Front)	1.035s
5-7	1.033s
10-12 (Center)	1.016s
15-17	0.991 s
19-21 (Back)	0.996s

Outcomes: The results showed that having a vertical COP closer to the centre of thrust is actually more beneficial, likely due to its role in minimising yaw.

Relevance to car: The findings from this test inspired the “shark fin” concept later on in development where a similiar fin is added on top of the canister chamber.



3.9: Shark fin concept
Credit: Lunar

Lift to Drag Ratio

Relevance to car: We aim for net zero lift and downforce to minimise drag and maximise canister efficiency. However, lift generating elements especially on the front wing is encouraged to help offset the tipping moment.

SUB-SYSTEMS RESEARCH

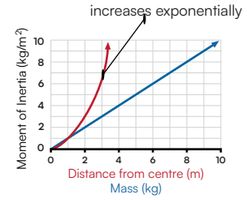
Wheel System Goals

Goal #1: Rolling Resistance

The rolling resistance coefficient of our wheels could be reduced in a number of ways:
- Low contact area with track.
- Minimise wheel deformation to lessen contact patch.
- Lessen friction coefficient of material.
- Smooth finishing on wheel to decrease friction.

Goal #2: Moment of Inertia

Moment of inertia is affected by the mass and the distribution of mass of the wheels. **Therefore, our wheels need to be as light as possible.**



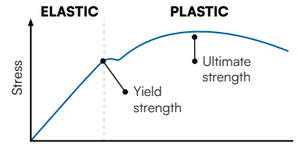
Goal #3: Stability

A stable wheel system is integral to consistent race results, loose spinning wheels waste large amount of energy, while a weak wheel supports make the car more likely to yaw.

Wheel System Material

Below is the 3 main points of deformation:
- **Yield strength:** The point at which deformation becomes permanent (cannot return to original shape).
- **Ultimate strength:** The maximum stress a material can handle before significantly weakening.
- **Fracture point:** The point where the material breaks or cracks.

Our wheels’ materials need to prevent entering the plastic region (hitting the yield point), and thus need strong tensile and compressive strength.



4.1: Material deformation graph
Credit: Lunar

Material Selection Chart

MATERIAL	DENSITY (g/cm³)	TENSILE STRENGTH (MPa)	(MPa)	MANUFACTURING METHOD	COST material + manufacturing (aud)
		Upon yield	Compressive strength		
ABS	1.1	34	65	CNC turning	25 + 250
Acetal	1.4	48	85	CNC turning	50 + 250
Nylon 12	0.95	30	53	SLS printing	200 + 200
PEEK	1.3	115	140	Injection moulding	250+5000
Polypropylene	0.9	33	40	Injection moulding	0 + 5000
Resin	1	58	63	SLA printing	14 + 0
HDPE	1	24	20	Injection moulding	9 + 5000

WHEEL SYSTEM DEVELOPMENT

WHEEL DEVELOPMENT

Aim: Find the wheel design that produces the quickest track times.

Wheel FEA Testing

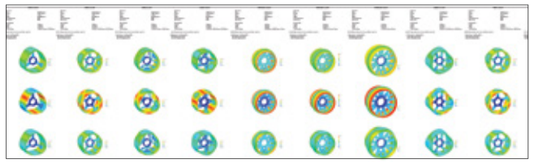
We used Autodesk Fusion's Static Stress Simulation for our FEA, as our wheels were already, so we can very easily import and analyse designs accurately and efficiently. These tests allow us to investigate the physical limits of different designs with different materials.

FEA Method:

- Ensure that the study materials are accurate (we tested all wheels with Polypropylene, and bearings with steel)
- Set a fixed structural constraint on the outside face of the bearing
- Set a pressure structural load on the outer rim of the wheel of 0.323MPa for rear wheels (1.5x value of total pressure on rear wheel (0.215MPa) calculated in Ansys Discovery), or 0.416MPa for front wheels
- Set a force structural load of 75N (1.5x value of estimated weight force on the car (50N)) onto the wheel, just below the bearing
- Set an angular global load about the centre of the wheel, at 1500 rad/s (same as CFD)
- Set mesh fidelity to 3% (or less if meshing is too large for the wheel)
- Generate mesh and solve simulation

Wheel Prototyping

We designed and tested a total of 11 sets of unique wheel concepts to cross-evaluate and compare to find the best solution.

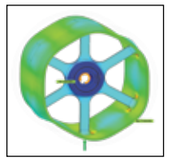


4.3: Extract from spreadsheet of all wheel prototypes, showcasing all properties of the wheel (area, density, mass, volume, size, FEA analysis)
Credit: Lunar

Standout Designs

**All wheel images (FEA) have an amplified deformation to highlight the weaker parts of the wheel more prominently.*

Six Spokes Concept



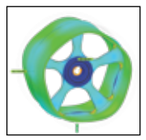
Mass: 0.760g
Moment of inertia: 135.235g/mm²
Stress: 17.594MPa
Displacement: 0.126mm
Strain: 0.022

Pressure on spokes, need larger fillets

Evaluation: Design has lowest moment of inertia and mass, it performs, however even spoke count leaves the wheel unsupported when the pressure is not on a spoke.

Improvement Actions: Add larger fillets to outer spoke edges to help dissipate stress. Add supports between spokes to help when pressure is not on a major spoke.

Five Spokes Concept



Mass: 0.882g
Moment of inertia: 155.732g/mm²
Stress: 12.796MPa
Displacement: 0.093mm
Strain: 0.013

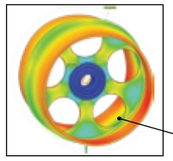
Evaluation: Simple and efficient, despite stress and strain appearing high they are mitigated due to its supports, however, it has a relatively high rotational intetia.

Improvement Actions: Reduce the mass of the outer spoke edges in order to reduce the rotational inertia. This lead to our "light five spoke" concept.



4.2: Light five spoke concept
Credit: Lunar

Five Holes Concept



Mass: 0.922g
Moment of inertia: 160.421g/mm²
Stress: 12.863MPa
Displacement: 0.101mm
Strain: 0.013

Extremely low stress and strain

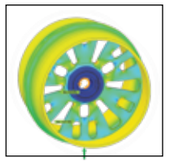
Evaluation: Extremely well performing in tests, however very heavy with high moment of intertia.

Improvement Actions: Experiment with more holes, otherwise spokes are the better option.



4.2: More holes concept
Credit: Lunar

Topological Optimisation Concept



Mass: 0.856g
Moment of inertia: 146.201g/mm²
Stress: 11.677MPa
Displacement: 0.100mm
Strain: 0.014

4.2: Manufacturability issues
Credit: Lunar

Evaluation: Therotically best combination between strength and weight, however difficult to manufacture due to close points of stress.



Improvement Actions: Increase the thickness of the centre and make smaller holes with more supports.



1st iteration
Displacement: 0.095mm
Strain: 0.012
Moment of inertia: 187.405g/mm²



2nd iteration
Displacement: 0.097mm
Strain: 0.013
Moment of inertia: 163.859g/mm²

Whilst these wheels improved strength, the increase in MOI was too much, so the initial wheel was the best design.

Wheel Design Results

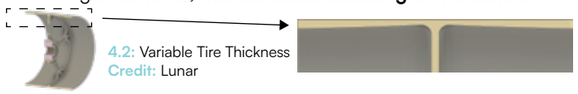
The table below is a collation of the data collected from each wheel's FEA test. For each of the five tests we took the wheel's value divided by the best value in the dataset. The percentages were added up to give a total score which ranked the wheels.

WHEEL DESIGN	MASS	%	MOMENT OF INERTIA	%	STRESS	%	DISPLACEMENT	%	STRAIN	%	TOTAL SCORE	FINAL RANK
Topological Optimisation	0.856	13	146.201	8	11.677	0	0.100	8	0.014	8	36	1
5 Spokes	0.882	16	155.732	15	12.796	10	0.093	0	0.013	0	41	2
5 Holes	0.922	21	160.421	19	12.863	10	0.101	9	0.013	0	59	3
6 Spokes	0.760	0	135.235	0	17.594	51	0.126	35	0.022	69	155	4

Evaluation: The topological optimisation was the best theoretical design, but due to the increased difficulty of manufacturing, we decided to select the **5 spoke design**, which was next best.

Variable Tire Thickness

Our simulations showed that the wheel edges deformed the most. To save weight, we could remove mass at the centre and add some to the edge. **However, this increased buckling at the centre.**



4.2: Variable Tire Thickness
Credit: Lunar

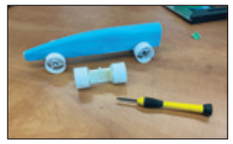
WHEEL SYSYEM DEVELOPMENT

Aim: Find the wheel system that produces the quickest and most consistent track times for stability.

Modular Wheel System Testing Rig

**Physical testing*

We developed a basic car which allows us to focus solely on the wheel system removing all other variables to do with car design. It has 3d printed attachments that use a simple screw in/screw out mechanism to easily interchange wheel systems.





4.2: Modular Wheel System Testing Rig
Credit: Lunar

Wheel System Designs

**Track Session #2*

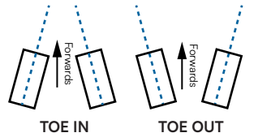


Insert Axle and Bearing | Insert Axle Stub | Insert Axle Continuous

Wheel Alignment Concept

**Track Session #2*

Adjusting the toe of the wheels can affect the stability. Having toe in decreases the yaw on the car, whilst toe out reduces the time to return the car to the centre of the track.



TOE IN | TOE OUT

4.2: Birds-eye view of toe in vs toe out
Credit: Lunar

Double Bearing Concept

**Track Session #2*

Hypothesis: Having two bearings in each wheel would provide additional stability, allowing the car to go straighter every time, reducing the race distance. However, extra mass affects the moment of inertia, but this is mitigated as the bearings sit close to the rotational axis.

Goal: To lower spin friction and increase stability.

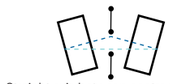
Wheel System Testing Results

We raced each system 20 times and took the fastest, slowest, and average time of each design. These were then ranked against the other wheel systems. The total score was derived from the sum of these separate ranks, the lower the better.

WHEEL SYSTEM	FASTEST TIME	RANK	SLOWEST TIME	RANK	AVERAGE TIME	RANK	SCORE	FINAL RANK
Double Toe (-Front +Rear)	0.939	1	1.016	1	0.974	1	3	1
Double Bearing	0.939	1	1.049	3	0.994	4	8	2
Double Toe Reverse (+Front -Rear)	0.956	4	1.119	7	0.987	2	13	3
+1.5° Toe In Front, Insert Axle Continuous Rear	0.969	5	1.052	4	0.999	5	14	4
1.5° Toe In	0.945	3	1.254	10	0.993	3	16	5
-1.5° Toe Out	0.980	6	1.068	5	1.010	6	17	6
Insert axle stub	1.007	8	1.040	2	1.022	7	17	6
-1.5° Toe Out Front, Insert Axle Continuous Rear	0.995	7	1.091	6	1.056	9	22	8
Insert axle and bearing	1.019	9	1.140	8	1.047	8	25	9
Insert axle continuous	1.032	10	1.140	8	1.061	10	28	10

4.2: Wheel system test results
Credit: Lunar

Evaluation: The toe out at the front and toe out at the rear was the best system, ranking #1 in all sectors. However, questions were later raised about how we would define the axis positions that determine the placement of Reference Plane A and B. As it is more important to be 100% compliant, we opted for the **double-bearing wheel** as it was the next best system that we had.



4.2: Toe axle definition issues
Credit: Lunar

BEARINGS

Bearing Selection

BEARING	DURATION OF SPIN (s)	TEMPERATURE (°C)	WEIGHT (g)
Hybrid ABEC 7	55	26.3	0.43
Full Ceramic	50	24.9	0.32
Solid-Steel	28	28.7	0.58

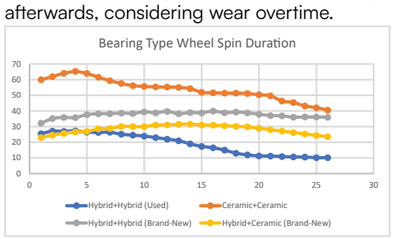
= Chosen bearing

Due to the similarity, we also have hybrid bearings as a backup.

Burnishing

**Physical testing*

To test the optimal wear of bearings to ensure they spin the longest, we used both a padded dremel and airbrush to control the spinning of our wheels (wearing of bearings), Spinning each for 5 seconds using the tool, and observing how long it spun for afterwards, considering wear overtime.



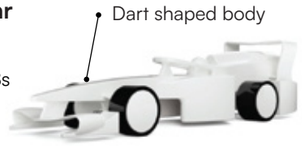


Conclusion: Ceramic bearing were most consistent. For spare wheels, this was our preferred option even through ceramic is extremely hard, which makes it resistance to wear. These bearings have an optimal window of approximately 7 races.

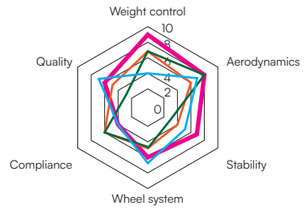
CAR DEVELOPMENT: PRELIMINARY DESIGN

DEVELOPMENT GROUNDWORK

Review of Nationals Car



Race time: 1.119s
Math model race time: 1.128s
Drag: 0.3182N
Lift: -0.0671N



= National Finals car
= State Finals car
= National finals car (2023)
= State Finals car (2022)

Improvement actions

Front wing: Integrate ramps in front of the wheel into the wingspan.
Body: Lengthen the canister chamber so that the transition from the back face of the halo to the round chamber shape can be stretched to be more gradual and occur over a further distance.

New World Finals Regulations

Possibilities: Multi element front and rear wings opens up new possibilities for wing design.

- Material change (Balsa to Foam)
- Addition of multi vane wingspans
- Addition of spare parts that we are allowed to subm

Limitations: The world finals requires a larger car to be built which, in addition foam blocks makes it harder to control weight, therefore our design has to be as lighweight as possible.

- Increase in Virtual Cargo size
- Increase in Exclusion Zone sizes

Expected Performance Drop:
Due to the increased size.

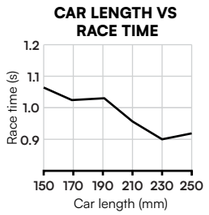
Aim: To determine the layout of the most vital components of the car (e.g. wheelbase length) in a way that optimises performance, these figures are then taken as parameters for our car design.

Car Length (KPI: 230mm)

**Track Session #2*

Hypothesis: A longer car helps to reduce pressure drag by allowing for more room to create streamlined shapes, whilst also being able to provide more stability to the car if weight is more spread out.

Method: A length adjustable 3D printed test car (65.4g) was created. The body of the car is broken down in modules being able to extend the cars length from 150mm to 250 mm. When modules were removed, weight was brought back up with screws.



Outcome: A longer car length is proven to be beneficial for the performance of the car, hence we decided to create our car with the longest possible length (250mm) allowed in the constraints of the block and regulations.

Multipurpose Testing Rig

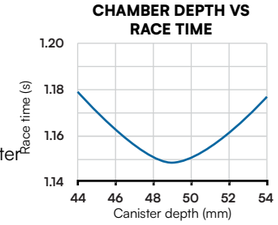
The following tests were done with version 2.0 of our multipurpose testing car, version one of this we used for our COG testing earlier on in the development process. Weighing 100 grams, this rig is designed with conducting multiple tests in mind.



Canister Chamber Depth (KPI: 47mm) **Track Session #2*

Hypothesis: It was thought that a longer canister chamber would help canister efficiency similar to the 'boat tail' concept.

Method: The canister chamber was intially designed to be 60mm deep. After testing the previous depth a two centimeter thick foam block into the the canister chamber to shorten its depth.



Outcomes: A deeper canister chamber was shown to have a negative impact on results, likely because it shifts the center of thrust forwards. 46 and 48mm produced the best and most consistent results hence 47mm was chosen as the sweet spot for our canister chamber depth.

Canister Chamber Tilt (KPI: 0° tilt) **Track Session #2*

Hypothesis: It was thought from our national's car that a 2° tilt is able to effectively offset the center of thrust misalignment caused by the tipping moment.

Method: The multipurpose test car was once again used in this test, and a 3D printed wedge slotted at the end of the canister chamber to shift is position up and down with a ragning from -3° to 3° degrees.

Outcomes: While a 2° tilt created the quickest single lap race time a neutral tilt proved to be more consistent so we ended up using it for our car.

TILT	TRIAL 1	TRIAL 2	TRIAL 3	AVERAGE
3°	1.597	1.591	1.586	1.591
2°	1.548	1.533	1.563	1.548
1	1.519	1.52	1.532	1.524
Netural Tilt	1.48	1.483	1.478	1.48
-1°	1.492	1.502	1.491	1.48
-2°	1.575	1.569	1.583	1.495
-3	1.598	1.612	1.597	1.576

Wheelbase Length (KPI: 130 mm) **Track Session #2*

Hypothesis: It was assumed that a longer wheelbase would create a faster car by providing greater stability.

Method: The test car had four axle holes to allow for an adjustable wheelbase length from 110 to 140 mm. The car was ran for three times under each configeration and the results are as shown below.

LENGTH	TRIAL 1	TRIAL 2	TRIAL 3	AVERAGE
110	1.555	1.512	1.543	1.537
120	1.482	1.486	1.507	1.492
130	1.423	1.439	1.455	1.439
140	1.413	1.447	1.421	1.427



Outcomes: While the longest wheelbase (140mm) did prove to have the best results, it did not have a significant difference when compared with 130mm. As 130mm allows for a longer front and rear overhang, we chose 130mm for our car to offer more tolerance and create more oppertunities for aerodynamic design.

CHASIS DESIGN

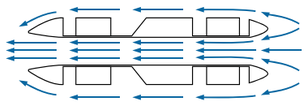
Goal: To create a body shape under the current that is able to direct airflow around the car in the most efficient way possible.

Car Body Concepts

The aerodynamic chasis of the car determines the general shape of the car moving forwards and has the greatest impact on the car's aerodynamic design. Therefore, it is important for us to choose the best car body shape before detailed development.

Catamaran Concept

Idea: This concept take inspration from real life catamarans to enclose both sides of the wheels in bodywork maximising laminar flow.



INSPIRATION:



Two teardrop shaped bodies

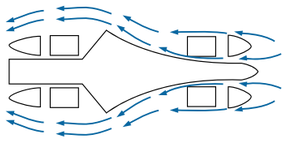
Drag Force: 0.184 N

Evaluation: The catamaran concept is extremely aerodynamically efficient, and by far the most aerodymaic chasis layout of all. However, it also requires the most amount of bodywork, making it challenging for weight distribution and management.

Dart Concept

Idea: This concept is ideal for weight management due to its low mass and narrow nose with minimal surface area. This was done on our nationals car, the Buzz v5.

INSPIRATION:





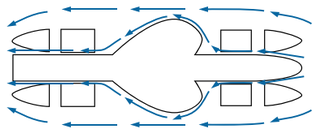
Drag Force: 0.199 N

Evaluation: While the sharp nose also help to minimise the stagnation pressure of air infront of the car, however this car has quite a high drag force.

Teardrop Concpet

Idea: The "open" concept aims to mimic the aerodynamic features of real life Formula 1 cars, the aim of car is to direct airflow under the rear wheels in the through teardrop shaped side pods.

INSPIRATION:



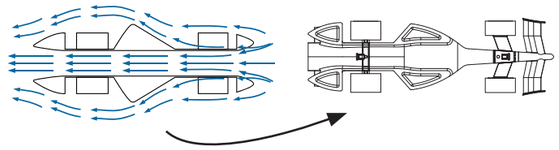


Drag Force: 0.211 N

Evaluation: This concept was not worth pursuiting as it causes too much drag.

Hybrid Concept *Chosen Concept*

Idea: The hybrid concept combines a dart shaped nose with a catamaran body, shifting the center of mass closer to the rear whilst maintaining a largely catamaran shape.



Drag Force: 0.184 N

Evaluation: This concept combines the best of both worlds beteween the catamaran and dart concepts, however due to the increased drag, it was not intially chosen. When we later struggled with weight issues, this concept was used for the final car.

Side Profile Concepts

Halo Placement (KPI: 20mm from canister chamber)

Four posistions (20mm, 40mm & 60mm) for the halo was tested in CFD to find the one that creates the least drag.

Outcomes: The rearmost position was choosen not only for the low drag that it produces, but also for its ability to bring the center of mass to the rear.

DISTANCE FROM CHAMBER (mm)	DRAG (N)
0	0.113
20	0.096
40	0.117
60	0.105

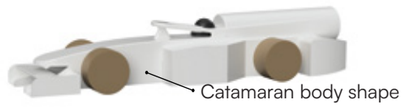
'STOCK CAR' PROTOTYPE

Goal: To create an intial prototype that combines all results from preliminary design to serve as the basis for further development.

Estimated Time: 1.061 s

Drag Force: 0.233 N

Lift Force: -0.0484 N



The "stock car" combined our findings from preliminary design regarding car length, wheelbase and chasis to create an intial prototype that serves as the 'platform' for our iterative development process of the car's components in detailed design. The car has minimal components to provide maximum felxibility for changes to be made later on.

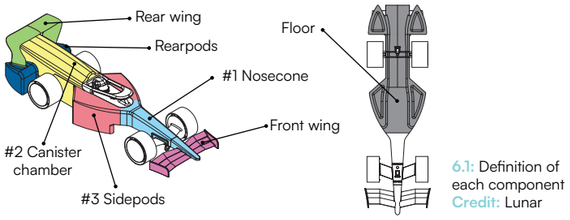
Evaluation & Improvement Actions

The "stock car" combined our findings from preliminary design regarding car length, wheelbase and chasis to create an intial prototype that serves as the 'platform' for our iterative development process of the car's components in detailed design. The car has minimal components to provide maximum felxibility for changes to be made later on.

CAR DEVELOPMENT: DETAILED DESIGN

Iterative Development Process

Goal: After defining the general shape of the vehicle, detailed design focuses on optimising each component individually.



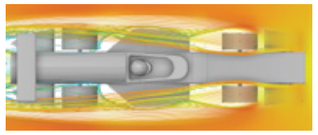
6.1: Definition of each component
Credit: Lunar

NOSECONE

Concept 1: Square

Idea: Controls airflow manipulation, split between top and bottom channels with clear underbody channel facilitating laminar flow.



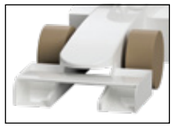


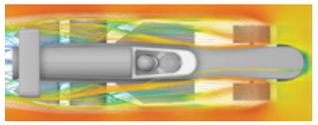
Improvement: Added weight and surface area adds resistance force, limiting speed.

Drag Force: 0.291 N

Concept 2: 'Duck Bill'

Idea: Splits flow between top and bottom, allows catamaran underbody channel easing flow smooth laterally from front edge. Tapered design smoothens transition at first contact with air.



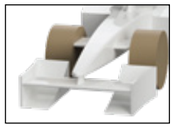


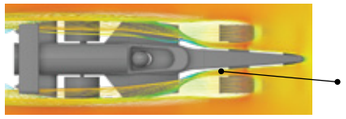
Improvement: More resistance created from being heavier and having more surface area.

Drag Force: 0.278 N

Concept 3: Narrow

Idea: Minimise surface area utilising light and sleek design focused on manipulating the little flow that attaches early laterally.





Drag Force: 0.280 N

Most streamlined airflow over wheels.

Additional concept: Dimpled nose

Idea: A dimpled nosecone aims to uitilise the surface area of the nosecone to create more laminar flow for incomming air, whilst directing airflow over and around the halo.

Evaluation: While this was a good aerodynamic concept, it caused manufacturing problems when the front wing is split up from the body.



6.2: Boat tail concept
Credit: Lunar

Final Nosecone

We decided on the **duck bill** for our final car, as it has the lowest drag and works best with the catamaran concept. This also allows us the most flexibility when developing our new components.

Improvement actions: Create steeper attack angle on the nose, helping direct airflow over or around the halo, bypassing halo's support structure on the centre.



CANISTER CHAMBER

Concept 1: 'Round'

Idea: Simplest solution allowing streamlined flow to easily attach to chamber are and towards the canister.

Improvement: Create more seamless transition to halo, improve aerodynamic efficiency by shrinking face.

Drag Force: 0.290 N

Concept 2: Helmet Transition

Idea: Aid streamlined flow attaching to chamber, tapered front follows helmet shape. Minimise halo's wake creating more laminar flow behind.



Improvement: Lengthen transition from halo and helmet to round chamber.



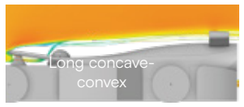
Drag Force: 0.267 N

Concept 3: Long Concave-Convex

Idea: Maximise catch of airflow early allowing it to continue to the canister and span a longer distance, creating more laminar flow.



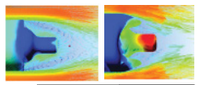
Improvement: Reduce mix of halo's wake and laminar flow, aerodynamics reduced due to proximity to nose.



Drag Force: 0.292 N

Additional concept: Boat tail

Idea: To sit on the back of the canister chamber and allow airflow passing over it to attach and follow its shape. The shape directs flow towards the back of the canister releasing the gas. This can create a momentary seal around the CO_2 stream, packing the gas tighter and increasing stream efficiency.



	DRAG (N)
No boat tail	0.3125
Body design	0.3027

Evaluation: While this concept did prove to be effective in CFD it pushes the limit of the regulations, and it was ultimitaly not worth the risk. However, we did continue with it for the detailed design prototype to continure exploring this concept.

Additional concept: Shark fin

Idea: Based on the findings from the centre of pressure (COP) test, we took inseparation fron 'shark fins' from the 2010s era formula one cars. This design shifts the center of pressure and improves the directional stability of the car by reducing yaw.



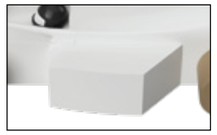


Evaluation: It was later learnt that while a shark fin is effective in mitigating yaw, but it also has the potential to amplify already exisiting yaw.

SIDEPODS

Concept 1: Rectangular

Idea: Start with a simple and easy to manufacture design to build upon.



Improvement: Don't trap wheel wake on the front face, and aim to direct airflow around rear wheel better.

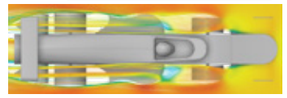
Drag Force: 0.286 N

Concept 2: 'Flick'

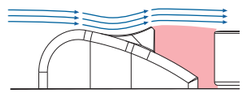
Idea: Move airflow around rear wheel and create an adverse pressure gradient to block the exclusion zone area.



Improvement: Make it easier to manufacture and reduce total surface area to minimise drag.



Drag Force: 0.292 N



= Adverse pressure
= Direction of airflow

Concept 3: Tapered

Idea: Increase canister efficiency by directing air between rear wheel and body, and catch untouched airflow as opposed to manipulating wheel wake.



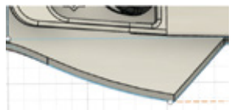
Improvement: Reduce airflow hitting the rear wheel.



Drag Force: 0.340 N

Final Sidepods

While the retangular sidpods was effective in maintaining the catamaran body shape and was the quickest of the three sidepods. It was not able to effectively manage the wake caused by the front wheel. Therefore, we went for a 'triangle' design for the final car.



REARPODS

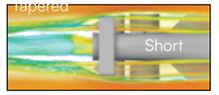
Concept 1: Diffuser shape

Idea: Squared shape shifts COG further back, closer to line of thrust. Back angled upwards and inwards, directing airflow towards canisters' CO_2 stream, increasing efficiency.

Improvement: The upwards diffuser shape was simply not effective.

Drag Force: 0.304 N



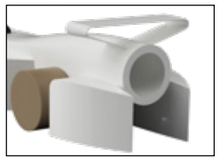


Concept 2: Kamm tail

Idea: Maximise airflow attachment, directing it inwards towards canister's CO_2 stream. Pressurises air, minimising stream moving in directions other than straight.

Improvement: The airflow was joined too upward from and the wake was not effectively minimised.

Drag Force: 0.281 N





Concept 3: Teardrop

Aim: Bring attached airflow on rear pod's outside to smallest point. Top edge taper to short edge reduces wake (turbulent flow) behind body. Theortically leaving the least wake.

Improvement: Added weight and surface area adds resistance force, limiting speed.

Drag Force: 0.278 N



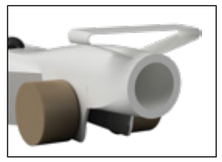


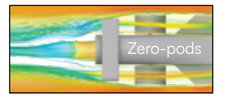
Concept 4: Zero Pods

Aim: Determine if rear pods are benificial. Manipulating wake from rear wheel may cause more drag than just allowing dirty flow to pass and join air behind the car.

Improvement: The zero pod design was suprisingly low drag, however having rear pods completes the catamaran design and shifts the COG rearwards.

Drag Force: 0.279 N





Final Rearpods

Low rearpods were chosen for having the lowest drag out of all options. The two other rearpods were high enough to block some air passing by as opposed to redirecting it. The zero pod design had the second-lowest drag, and would be used if the final car was overweight as it reduces the amount of foam. However, looking at the CFD capture, the air is far more turbulent than those with rear pods, potentially slowing down the release of air from the canister, which is more harmful than 1 gram of weight.

CAR DEVELOPMENT: DETAILED DESIGN

REAR WING

Concept 1: Curved

Idea: Shift COG closer to car's rear, aligning closer to line of thrust. Increase energy efficiency through vertical supports increasing mass, contributing minimally to car's overall frontal area.

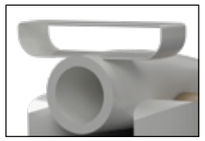


Drag Force: 0.265 N



Concept 2: Box

Idea: Design with smallest frontal area. Loosely curved side dropped supports create larger separation between top, sides, and bottom for more passing, uninterrupted laminar flow.



Drag Force: 0.299 N

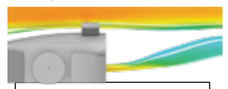


Concept 3: Triangle

Idea: Minimise rear wing's frontal area. Two angled supports underneath creates dihedral shape, increasing lateral stability, preventing roll caused by irregular canister puncture.



Drag Force: 0.279 N

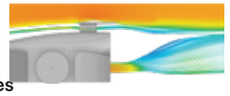


Concept 4: Delta Wing

Idea: To maintain a very minimal frontal area, but also uses a swept shape. This shape increases the longitudinal stability, minimising energy loss from launch. Unfortunately it uses a single central support that is deemed illegal by the regulations.



Drag Force: 0.306 N





»» Additional concept: Dihedral angles

Upward wing angles are used on planes to increase lateral stability. It could do the same for cars, but if glued out of place, it could end up causing more roll instead of less, and was thus too risky.

»» Additional concept: Weighted rear wing

A concept that was thought of was to create weighted rear wing to bring the center of mass furthur upwards above the canister chamber. However, this was not used as our car ending up struggling in loosing weight.

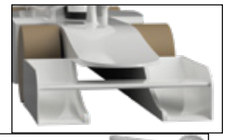
Final rear wing

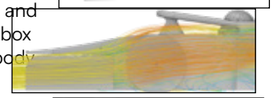
We chose the **curved design** as it had the lowest drag force by a significant margin. It is estimated that this is due to the vertical support structure, only affecting the already turbulent sides of the car's drag compared to the laminar centre.

FRONT WING

Concpet 1: Boxed

Idea: Simplicity via low profile neutral wingspan, wheel ramps, and endplates for lateral flow. Open box for laminar airflow in the underbody.



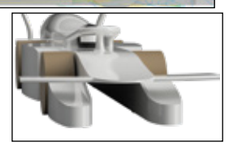


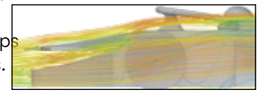
Drag Force: 0.291 N

Lift Force: 0.294 N

Concept 2: Hammerhead

Idea: Wingspan extends from nosecone, minimizing frontal area and avoiding extra support structures. Boat hull-shaped ramps push airflow laterally over wheels.





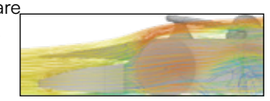
Drag Force: 0.213 N

Lift Force: 0.193 N

Concept 3: Catamaran

Idea: Fit two span segments within same frontal area. Provides greater strength to wheel ramps, which are now concave and lower profiled.



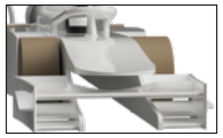


Drag Force: 0.274 N

Lift Force: 0.107 N

Concept 4: Multi vane

Idea: Multi-element sections manipulate flow with less frontal area. Boxed shape and canards maintain laminar airflow direction. Ramps push air over wheels.



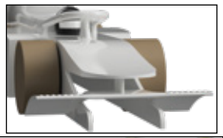


Drag Force: 0.259 N

Lift Force: 0.0496 N

Concept 5: Lift

Idea: Generates lift to slightly raise front wheels, reducing friction. Vortex generators reduce airflow separation, aligning boundary layer with wing shape.





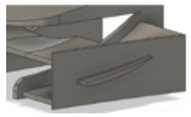
Drag Force: 0.286 N

Lift Force: 0.0527 N

»» Additional Concept: Cannards

Inspired by car design design, cannards direct air parallel with the outside of the wheels and try to scoop the air over the wheels whilst generating minor vortexes.





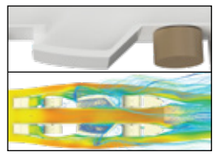
Final front wing

The box front wing was chosen for the final front wing desing due to its low drag, simplicity and effectiveness in directing airflow over and around the wheels.

FLOOR MODIFICATIONS

Concept 1: Sidepod duct

Idea: To redirect previously turbulent air in front of the sidepods and to bring them into the laminar floor chamber.



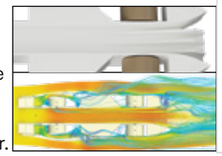
Drag Force: 0.313 N

Lift Force: 0.171 N

7.12: Floor iteration 1
Credit: Lunar

Concept 2: Diffuser

Idea: to reduce the adverse pressure gradient created between the high velocity, low pressure floor tunnel and the medium velocity and pressure rear of the car, enabling a smooth transition, reducing vortex generation at the canister.



Drag Force: 0.274 N

Lift Force: 0.102 N

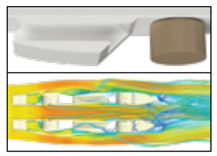
*Inspriation



7.13: Floor iteration 2
Credit: Lunar

Concept 3: Sidepod cutout

Idea: Similar to the sidepod duct, to remove the turbulent air buildup behind the front wheels without disturbing the laminar flow in the centre of the floor.



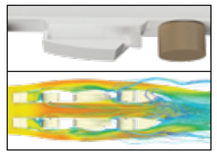
Drag Force: 0.255 N

Lift Force: 0.0413 N

7.14: Floor iteration 3
Credit: Lunar

Concept 4: Sidepod outlet

Idea: To redirect the turbulent airflow at the front of the sidepods over the wheels as opposed to the sides.



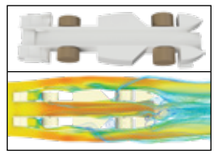
Drag Force: 0.266 N

Lift Force: 0.0281 N

7.15: Floor iteration 4
Credit: Lunar

Concept 5: Simple

Idea: To have a simple design with less places for aerodynamic issues.



Drag Force: 0.259 N

Lift Force: 0.0496 N

7.16: Floor iteration 5
Credit: Lunar

Final Floor Modifications

The sidepod cutout had the lowest drag but only slightly outperformed the simple design, which lacked front wheel turbulence fixes. The simple design, with a filleted front edge, would be quicker. CFD did not show airflow issues affecting the rear wheels or causing excessive lift and nosedive. Thus, the simple floor was chosen for reliability.

»» Improvement actions

Use above designs if there are specific issues with the car, such as increasing lift with sidepod ducts. Can also hollow out sidepods and rearpods to save weight or adjust CoG.



7.16: FHolled out sidepod
Credit: Lunar

DETAILED DESIGN REVIEW

Final Prototype

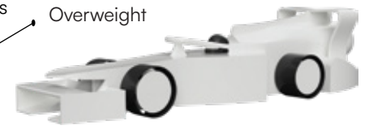
Estimated Time: 1.061 s

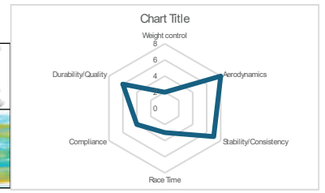
Track Time: 1.211 s

Drag Force: 0.233 N

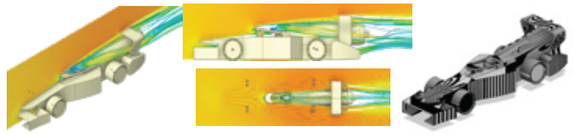
Lift Force: -0.0484 N

Overweight





The detailed design car was extremely aerodynamically efficient with advanced aerodynamic features such as the shark fin and the boat tail. However, it struggled with weight due to its large size created by the catamaran body.



»» Review Meeting

A meeting was conducted with a the engineering team and various mentors including a former engineering judge to find flaws in the current design. Key results from this meeting include creating a simpler design in favor of compliance, and considering generating lift with the car to minimise the tipping moment and wheel friction. The car's design was also too heavy.



Boat tail

»» Improvement Actions

Create Simpler Design: The design of the car way very complex making certain elements unpredictable.
Revert to dart shaped nose: The
Regulation Compliance: Elements such as the 'shark fin' and the 'boat tail' pushes the limits of the regulations.
Weight issues: The catamaran body shape of the car caused the final manufactured product to be extremely overweight at 67.7 grams, hence a lighter design is needed.
Regulation Compliance: The car design broke regulations in 8 places.





67.7 grams

»» Goals For New Car

Hybrid Body shape: The final car would change from the catamaran body shape to the 'dart' shape in favor of weight saving.
Lift Generating Wing: The final car should consider using a lift generating or multi vane wing to offset the tipping moment and lesses wheel friction.
Aerodynamic Simplicity: The final car should focus on creating 'simple yet effective' aerodynamic elements as complex aerodynamic components can be deliver inconsistent results due to the small scale of the car.

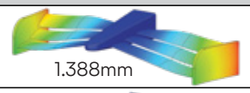
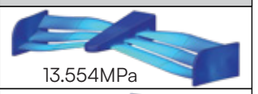
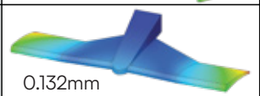
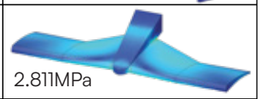
CAR DEVELOPMENT: REFINEMENT & PRODUCT

COMPONENT REFINEMENTS

Front Wing

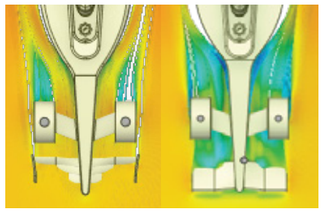
Aim: To find the fastest and most durable front wing. We tried a lift wing to see if raising wheels can reduce friction.

We tested the strength of the wings using FEA and the lift wing was far stronger than the multi vane. However, the multi vane was still strong enough to pass our safety standards.

	DISPLACEMENT	STRESS
MULTI VANE	 1.388mm	 13.554MPa
LIFT	 0.132mm	 2.811MPa

8.1: Refinement front wing FEA
Credit: Lunar

8.2: Refinement front wing CFD
Credit: Lunar



The lift car blocks off the airflow, causing significant wheel wake in front and behind the wheel. Also, when the air rejoins at the sidepods, because the multi vane keeps its speed, the pressure gradient is reduced, minimising flow separation.

MULTI VANE	VS	LIFT
0.252 N	Drag Force	0.271 N
0.32	Drag Coefficient	0.37
0.0416 N	Lift Force	0.0819 N

When we track tested the two cars, the multi vane was, on average, 0.007s faster, despite having 0.1g less air, estimated to be worth an additional 0.008s.

	MEDIAN TIME	CANISTER AIR
MULTI VANE	1.071s	7.6g
LIFT	1.078s	7.7g

Evaluation: Despite being structurally weaker, the **multi vane front wing was chosen** due to its better aerodynamics and faster track test lap times. It also helps reduce mass and tipping moment.



8.3: Front wing track testing
Credit: Lunar

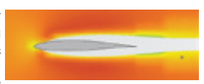
Rear Wing

Aerofoil Design

Aim: To find the aerofoil with the least effect on the car

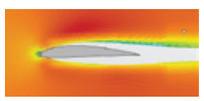


NACA0012
Neutral aerofoil, simple, easy to manufacture
Drag Force: 0.0086N
Lift Force: 0N



8.4: Rear wing aerofoils
Credit: Lunar

NACA63-210
Neutral Aerofoil, complex, difficult to manufacture
Drag Force: 0.0073N
Lift Force: 0.0004N

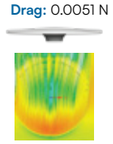
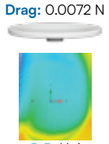
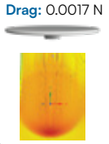
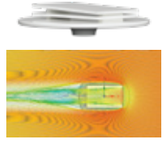


DAVIS
Lift profile, intricate, relatively easy to manufacture
Drag Force: 0.0079N
Lift Force: 0.0093N

Evaluation: Whilst NACA63-210 was the sleekest of them all, it would be too difficult to SLS print the trailing edge on such a small scale, so the **NACA0012 was chosen** instead.

Hubcaps

Aim: To reduce the turbulent air produced by the wheel, and generate velocity to speed up the car.

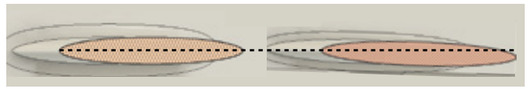
Convex	Concave	Flat	Vortex Generator
Drag: 0.0051 N	**Drag:** 0.0072 N	**Drag:** 0.0017 N	**Drag:** 0.0019 N
			

8.5: Hubcap designs
Credit: Lunar

Evaluation: The **flat hubcap** was the most drag-efficient item and was thus chosen for the final car. The simplistic design reduced space for aerodynamic issues, keeping the car streamlined.

Front axle attachment

Aim: To lift the front wheels up without affecting the drag force.



FLAT ATTACHMENT
Lift: 0.016 N

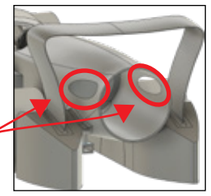
2° ANGLE
Lift: 0.0416 N

8.6: Front axle attachment angle of attack
Credit: Lunar

Evaluation: The 2° angle was chosen as it provides lift directly at the wheels and fits within the same frontal area, reducing wheel friction without having an impact on the rest of the car.

Compliance Refinements

One major issue was T5.5, the Chamber Safety Zone. We initially used three profiles (at the back, halo, and in front of the canister hole) with a 3.5mm offset, aiming to meet the 3mm regulation. However, loft transitions created grooves under 3mm. To fix this, we added a fourth profile at the problem area and increased the offset to 4mm.



8.7: T5.5 breach points
Credit: Lunar

CFD SETUP

Aim: To gather accurate aerodynamic performance metrics, including drag and lift, and to understand what makes the car fast/slow by analysing simulated wind tunnel images.

CFD Methodology

- In Fusion, add wheel patches and merge all components into one body
- After importing the file into Discovery, create an external flow setting the inlet as the front plane and the ground plane as the bottom of the car.
- Change the flow inlet speed to 20m/s
- Set the material of the fluid flow as "air", and the car as "ABS Plastic"
- Create rotating walls at 1500 rad/s (-1500 if not rotating forwards) for the 'tyre' on each wheel, and one more for all other components of the wheel (axis of rotation is slightly different due to flat patch on tyre)
- Create force monitors by selecting the whole car, and then choosing the axis that is going forwards for drag, and the vertical axis for lift/downforce
- Set calculation time to "time-dependent", specify calculation time as 0.5s, then put fidelity to 100%
- Simulate and record data from monitors and capture the simulated images on multiple angles

FINAL CAR DESIGN



8.8: Final Car
Credit: Lunar

Estimated Time: 1.065 s **Track Time:** 1.074 s
Stress: 13.554 MPa **Strain:** 0.018 **Displacement:** 1.388mm
Drag Force: 0.255 N **Drag Coefficient:** 0.32 **Lift Force:** 0.0416 N **Frontal Area:** 2278mm²

We analysed every key feature of Lightyear 1, discovering the direct changes that they make to the rest of the car, and the overall benefit that each component gave us with reference to our three engineering performance goals (weight, aerodynamics, stability). When combined, this gives us the best overall car design out of all of our different ideation processes and data-driven analysis techniques, justifying our decisions.

FEATURES	ADVANTAGES	BENEFITS
Multi Vane Front Wing **(WS)**	Moves CoG back	Reduces tipping moment for stability and overall weight
Narrow Nosecone **(WAS)**	Low frontal area	Lowers the drag coefficient, boosting efficiency
Lifting Aerofoil Front Axle Attachment **(WA)**	Reduces axle load and wheel friction with track surface	Lowers deceleration rate and negative forces on the car
Double Bearing Wheel System **(S)**	Enhances wheel rotation efficiency	Car travels in a straighter line, decreasing distance required
Interchangeable Wheel System **(S)**	Bearings can run at peak performance every race	Reduces friction in bearings, increasing efficiency
Front Tapered Sidepods **(AS)**	Increased velocity on inside of wheels	Lowers drag and improves wheel stability
Clear Air Pocket Around Rear Wheels **(S)**	Decreases air resistance, preventing drag build-up	Improves rear stability and efficiency
Helmet-to-Canister Chamber Transition **(A)**	Eliminates sharp airflow disruptions	Reduces drag

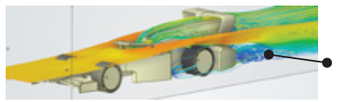
8.9: FAB Analysis of Final Car
Credit: Lunar

COMPUTER-AIDED ANALYSIS: CFD

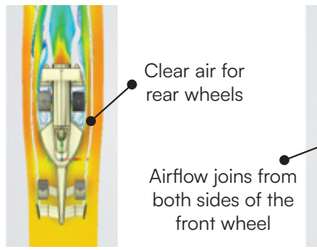


8.10: Final car CFD
Credit: Lunar

Clean join of air at back



Built-up air forces fast-velocity air from above to go towards the canister, helping direct the gas released



Clear air for rear wheels

Airflow joins from both sides of the front wheel

Reduced Wheel Wake

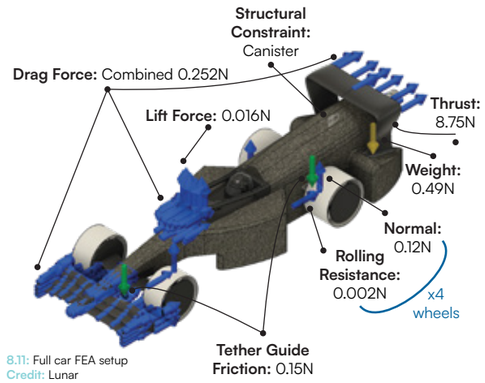
Physical Visual Analysis

The streamlines validate our CFD, that the only points of buildup are behind the car and at the helmet. It also demonstrates how the angled nosecone helps to direct airflow above the halo notch.

8.11: Final car physical visual analysis
Credit: Lunar

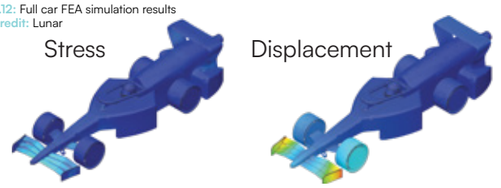


COMPUTER-AIDED ANALYSIS: FEA



8.11: Full car FEA setup
Credit: Lunar

8.12: Full car FEA simulation results
Credit: Lunar



3D MODELLING

CAD SETUP

Software - Autodesk Fusion 360

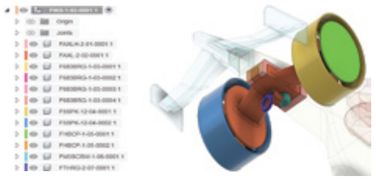
Autodesk Fusion 360 was our choice of CAD software because of its simple workflow, cloud based approach, and extensive workbenches that allow us to create complex surfaces and geometries. Being cloud based made it easy to collaborate and share files.

Organisation

When working with many parts in assemblies and multiple prototypes and iterations it can be difficult to access things quickly. We used a modified version of an existing file structure from Marand Precision Engineering, an Australian defence force company.

[Shorted Part Name]-[Iteration number]-[Sub Assembly number]-[Part number]

Example: RWNG-2-06-0001



Sketches

Hand-drawn sketches were crucial to a creative brainstorming process, visualising designs and work with surrounding components. Drawings were created for every idea before CAD modelling and organised in a OneDrive folder. This process is efficient, promoting easy thought collaboration.

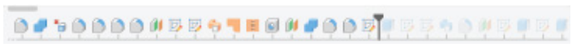


9.1: Hand-sketches of initial concepts
Credit: Lunar

MODELLING TECHNIQUES

Parametric Modelling

This design process allows easy modifications and adaptations of parts with little errors. This allowed us to change a certain dimension or shape for testing/legality reasons without restarting. On Fusion, turning on the timeline activates this ability. Using this throughout our development process determined general shapes and relative sizes of car components used for CFD, which we later refined. It sped up general development of modelling.



9.2: Image of timeline used for parametric feature
Credit: Lunar

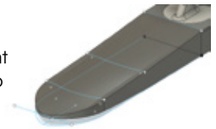
Surface/Solid Modelling Combination

Solid modelling

Solid modelling constructs bodies by adding and subtracting volumes known as boundary representation, created from closed sketches (faces). Solid modelling is quick and easy, creating easily identifiable shape properties.

Surface modelling

This consists of multiple profiles or faces creating more detailed shapes with extreme surface continuities that transition smoothly. Using patches to connect surfaces and a stitch to connect all of the surfaces and patches into a solid.

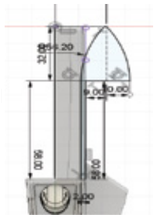


9.3: Image of Nosecone Loft - detailed surface continuity
Credit: Lunar

Our combined modelling approach cohesively reaped rewards from both workbenches. Solid modelling created rigid, simple, consistent shapes such as car wheel systems. They must be created to specific dimensions accommodating to external parts. Surface modelling was used for varying boundary curvature that would otherwise be too complex to create as a solid, like our car's canister chamber loft.

Sketch Constraints

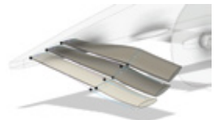
Sketches are the bases defining all shape creations. We must constrain them correctly to utilise parametric modelling with the timeline and remove accidental changes to sketches which create irregularities. In Fusion 360, our sketches were fully constrained, indicated by lines (originally blue) turning black in colour.



9.4: Image of master sketch - fully constrained
Credit: Lunar

3D Sketches

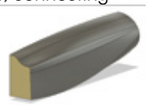
Rather than being attached to just one plane in a normal sketch (2D), 3D sketches act between multiple planes, yielding 3D characteristics. Rather than being used as profiles, they often work as guide rails in different functions, manipulating curves by pulling or pushing the profiles to match. We used 3D sketches when lofting or sweeping profiles. Our wingspans are aerofoils at the cross-section, but from a top plane's view they can be swept backwards whilst also tilting upwards from a front-on-view.



9.5: Image of 3D sketch shaping a loft/sweep - wingspan
Credit: Lunar

Lofts

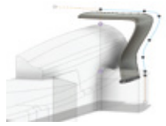
This creates a 3D solid or surface within the space, connecting several cross-sections to be manipulated by guide rails further pulling the profiles. We used a combination of solid and surface lofts. Our canister chamber is a surface loft going between 3 profiles, making sure it maintains regulatory chamber thickness, seamlessly transitioning to the halo's back.



9.6: Image of surface loft - Canister chamber
Credit: Lunar

Sweeps

Sweeps consist of a profile and a path at its core. They allow us to make intricate curvature that can maintain the same shape throughout its whole path. We used sweeps primarily for our front and rear wing spans so that we could have an aerofoil shape across all areas of the spans.



9.7: Image of multi-element wingspan sweep
Credit: Lunar

Surface Continuity



G0: Refers to two adjacent lines that sharply meet at and edge creating a non-radiused corner. This is the simplest form of CAD transition that is impossible to manufacture due to machining restraints of a ball nosed drill bit.



G1: Refers to a tangent transition between two adjacent lines. The surfaces share a common angle and are continuously tangent at their junction. This transition is created through simple fillets/radii.



G2: Refers to two lines that have an even tangency weight at the blend. It is flush throughout curvature, but difficult to consistently recreate. Usually created through free form modelling and is difficult to maintain consistently throughout edges.



G3: Refers to an advanced level of continuity with a consistent rate that it changes at on the surface. It is harmonious as both length and shape are smooth with little horizontal/vertical surfaces. This shape is difficult to create, leading to modelling errors from constraining challenges and is tough to machine cohesively.

9.8: G0-G3 surface continuity
Credit: Lunar

Curvature Comb/Isocurve Analysis

Curvature Comb Analysis consists of elongated lines that follow the curve of the selected geometry. This exaggerated view is useful for determining possible errors for intense shapes such as the sweep spline from our rear wing. If errors occur, the combs would overlap and display on top of each other as opposed to being smooth, tangent, and separate from each other.



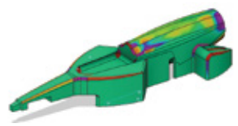
Zebra Analysis

Zebra Analysis places black and white stripes onto the car which allows us to visualize curvature on smooth surfaces. This was particularly useful for identifying the surface continuity for manufacturing, and to ensure that joins between parts remained even and followed the same curvature.



Curvature Map Analysis

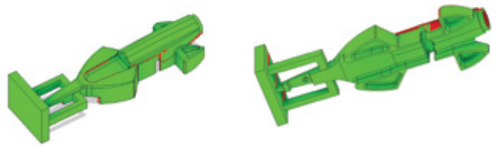
The Curvature Map Analysis displays a wide arrange of colours as a gradient on the selected surface. The deeper the colour shown on the body, the more intense the curvature is within that area. We aim to have majority of the surface shown as green to validate that the curvature is lengthened and smoother, making it easier to manufacture, and better for airflow staying attached.



DESIGN TO MANUFACTURE

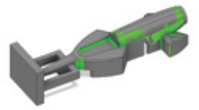
Accessiblity Analysis

Accessibility analysis colourises the model to show reachable areas of manufacturing and restricted areas too. We used this to check and see if the car could be manufactured top and bottom without needing any side-to-side milling.



Minimum Radius Analysis

When machining using a 3mm ball nose drill bit, we understand that the smallest radius we can create would be a 1.5mm. To ensure that all radius' would show up after machining, we made sure that everything showed up green. This is useful for ensuring that separated parts like the rear wing would still fit within pockets inside the body.

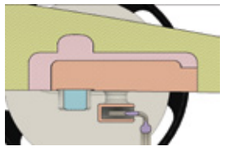


Billlet Support

Before we machine our cars we create tailored billet supports to our clamping system. With our design we are weak at the nose cone. To support this we created two arms that wrap around it and join at the start of the nose. This we ensured to have as much surface area as possible to make sure it was as strong as it could be without disrupting the design of the model.

Part Clearance

Adding clearance to parts allows them to fit together nicely. If no clearance is added, and the parts are supposed to fit to each other perfectly, due to our SLS machines tolerance of ±0.1mm and the CNC tolerance of ±0.05mm, parts often come out either too big or too small. To combat this, on all of our joins between SLS parts, we added a -0.1mm clearance to all faces, and for foam bodies we added a -0.05mm clearance.



CAD Tolerencing

We added tolerances that ranged from 1-3mm extra room which could allow for deviances between parts from machines (±0.05mm for our CNC router, and ±0.1mm for our outsourced SLS Nylon 12 parts). It also accounts for any possible manufacturing errors, like imperfect gluing, thicknesses from paint, and sanding. Tolerance allowed us to make a few minor mistakes, and still be within the ranges for regulations, leading back to our overarching goal of have a fully compliant car.

Radius/fillets

Radius's and fillets of different types were used on all edges of the car. This creates a CAD model most accurate to the final car which will be manufactured. This is because of our understanding of machining constraints due to using a ball nose drill bit. When using a 3mm bit, the smallest radius we can create is 1.5mm on foam parts.

Constant

Constant fillets remain a consistent tangent radius of G1 continuity throughout a curve.

Variable

Variable fillets, vary in sizes as they move along the edge. These vary through different surface continuity if we would like variation along the same edge.

USE OF CAM/CNC

MACHINING PREPARATIONS

Manufacturing Objectives

Overarching manufacturing objectives

- **Regulation compliance:** Ensure the final model is 100% compliance with all regulations.
- **Weight:** Be between 50.03-50.08 grams.
- **Stable wheel system:** Ensure our wheel system is manufactured as intended to maintain stability.

Sub goals

- **CAD accuracy:** Replicate CAD model as accurately as possible to achieve virtually tested results.
- **Minimise hand-finishing:** To reduce the change of inaccuracy/mistakes caused by hand work.
- **Smooth surfaces:** To minimise drag along body.

Order of importance

Weight Management

To ensure our car doesn't fall below 50g, we aimed for our car to weigh between 50.03-50.08 grams when first on track, because as previously witnessed, this weight may change due to external factors during the competition and travel. To ensure we achieve this goal, we considered the following factors that may impact the weight while travelling:

VARIABLE	RISK	AVG. WEIGHT LOSS (%)	AVG. WEIGHT LOSS (g)	REASONING
Moisture lost	Highest	-0.4	-0.20	Lower humidity and higher air temperature.
Off-gassing	Medium	-0.1	-0.05	During the period of paint drying, there is a natural release of gas (3-5 days).
Wearing	Very low	-0.02	-0.01	Wearing from non-dry and high contact handling of car.
TOTAL ADDED WEIGHT:			+0.26g	

10.1: Degradation of car mass, and added weight to counter
Credit: Lunar

Melbourne, Australia compared to Dammam.

To test this, we stored a 62.3g car in a humidity box, and measured the following after 72 hours.

TIME	WEIGHT (g)		HUMIDITY (%)	
Day 0	62.3		18	
Day 1	62.4	+0.1	18	
Day 2	62.56	+0.26	56	+38
Day 3	62.59	+0.29	55	+37

10.2: Weight change vs humidity test
Credit: Lunar

Conclusion: Humidity increase = weight increased (rate of increase slows overtime). Due to the decreased humidity in Dammam, we can assume our car after 72 hours would gain ≈0.3g, so we left a clearance of -0.4g.

Estimating block weight

To calculate which weight block to use in the CNC, we estimated the weight of our car by calculating the densities of our heaviest, average, and lightest block available.

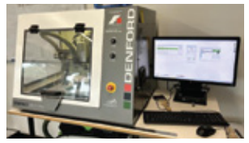
	HEAVIEST	AVERAGE	LIGHTEST
Volume (cm^3)	703.62	703.62	703.62
Block weight (g)	123.54	118.75	114.77
Density (g/cm^3)	0.176	0.168	0.163
Car weight (g)	21.30	20.33	18.47

10.3: Block weight calculations (from our range)
Credit: Lunar

This helped with weight calculations and spindle speed and feed rate calculations. We also used our 6 lightest blocks as our primary race quality cars in order to get the race cars foam as light as possible as it takes the most volume.

CNC Equipment

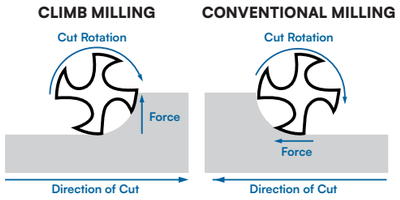
Our CNC has the ability to manufacture with live control of spindle speeds and feed rates, with 24,000RPM spindle speed and 5000mm/min feed rate, ensuring precision. This also included a clamping system for changes in billet orientation, locked in place on a bed volume of 400mm x 240mm x 110mm. We found that our spigot in our clamping system has a slight kink of an angle, decreasing machining accuracy. To solve this we made sure to always keep the same point of the spigot facing up.



10.4: Denford Compact 1000 Pro CNC
Credit: Lunar

Conventional vs climb milling

Climb milling: is when the spindle cuts in the same direction as the blocks feed which decreases chipping and smearing, like it's climbing the block.
Conventional milling: is when the spindle moves against the materials feed direction, which produces larger chippings and lowers surface quality.



10.5: Conventional vs climb milling
Credit: WayKen Rapid Manufacturing

We found that climb milling was a cleaner method but we chose to stay with bi-directional milling. This was because the quality between the 2 methods was different but similar, thus we chose the more time efficient option.

CAM Settings

Solving spindle speed & feed rate problems

First, we aimed for a 900 Surface Feet per Minute (SFM) value as this was within the typical range for foam, which is 800-1000 SFM. We then determined the spindle speed for our 3mm drill bit:

$$\text{Spindle speed (RPM)} = \frac{\text{Cutting speed (SFM)} \times 304.8}{\pi \times \text{Tool Diameter (mm)}}$$

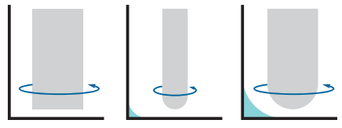
10.6: Spindle speed formula

$$\text{RPM} = \frac{800 \times 304.8}{\pi \times 3} = 25{,}872$$

However a problem was that the RPM cap for our machine was only 24,000, so we weren't able reach our SFM goal, meaning we could only use 742SFM which still cuts clean but is not fully efficent.

To solve this we used both our ball nose 6mm and 3mm drill bits, allowing us to mill the foam effectively with the precision advantage of the 3mm drill bit.

Drill bit: We used a ball-nose drill bit as the curved cutting surface is able to create fillets on our model, which isn't possible with a flat nose. We measured the diameter of the drill bit which was 5.96mm, Measured using digital callipers with a tolerance of ±0.01. Being less than 6mm meant that there would be an increase of material along more vertical faces, creating smoother surfaces.



10.7: Drill bit comparison
Credit: Lunar

= not accessible

We then calculated the spindle speed with the 6mm drill bit with 1000SFM:

$$\text{RPM} = \frac{1000 \times 304.8}{\pi \times 6} = 16{,}170$$

With Spindle Speeds of 24,000 and 16,170RPM we then used feed rates varying from 1,500-5,000mm/min, ignoring the set theoretical feed rate calculations because we cannot use just 1 speed for the whole car:

Roughing plans: 2,000mm/min (top), 1,300mm/min (bottom).

Refinement: 3,800mm/min (top) and 2,300mm/min (bottom).

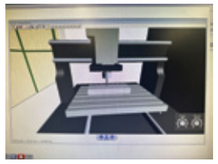
High finishing: 5,000mm/min for both sides.

By increasing feed rate as the amount of cutting reduced, smoother surfaces were achieved, and minimised hand-manufacturing.

CAM Software

To CNC our cars we used QuickCamPro to create g-code for the CNC to use and VR-Milling to control it. QuickCamPro allows us to create the g-code required to machine by inputting 6 key steps:

1. Orientation
2. Cut depth
3. Billet size
4. Machine plans
5. Preview
6. Datum point



10.8: VR milling software
Credit: Lunar

We would then transfer this code into VR-Milling, where we would be able to connect, control and simulate the machine. Whenever we were not sure in any milling code, we would run a digital VR version of our machine to check the code process was correct.

Machining Plan Problem Solving

When milling, sometimes our foam block twisted, causing inaccurate milling for the rest of job. To combat this, we started by milling the top half first to provide more strength when milling the bottom and our alteration in feed rates reducing the twist.



10.9: Block twisting problem
Credit: Lunar

Top milling:

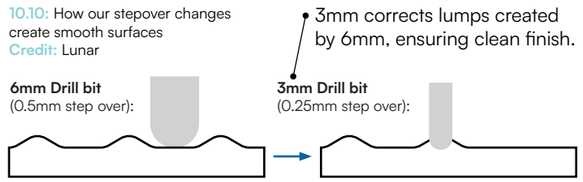
- Area clearance - to get rid of void material without yet milling the cars surface.
- Raster Roughing (0°) - removes more material closer to the car's surface.

These 2 plans to got rid of excess material, reducing the amount of cutting contact in the final finishing without reaching the models surface yet.

Finish:

- X2 Raster finishing (225°)
- X2 Raster finishing (315°)

} **0.5mm & 0.25mm stepover.**



10.10: How our stepover changes create smooth surfaces
Credit: Lunar

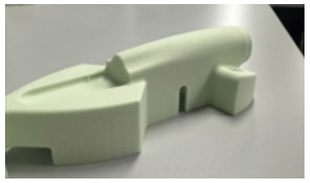
When milling the bottom half, we did the same process but removed the first area clearance because of the thinner and smaller walls which would tend to break easily.

ISSUES & PROBLEM SOLVING

Milling Problems

Machine offset

When milling, our CNC has an offset by machining the model slightly to the right on each orientation.



10.11: Offset problem on car
Credit: Lunar

Manufacturing Problems

Difficult to reach areas

So we could sand in difficult to reach areas (e.g. small crevices), we created custom printed tools to reach these areas.

We also recycled an electric toothbrush, and replaced the bristles with sandpaper disks, allowing for comfortable handling when sanding in difficult to reach areas.



10.12: Custom sanding tools
Credit: Lunar

These innovations also helped mitigate potential damage caused without them.

Sealant

During manufacturing of prototypes, we realised that foam absorbs spray putty and paints, increasing the weight of the car. To prevent this, we tried sealing the foam body with:

- Polyurethane
- Resin
- PVA glue



10.13: Sealant materials
Credit: Lunar

Each mixed with mater to not clog our airbrush. Each was able to seal the foam to prevent absorption, however the PVA glue solution provided the best surface finish.

OTHER MANUFACTURING AND ASSEMBLY

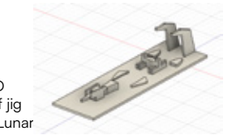
ALL MANUFACTURING STAGES

Sealing

We started by airbrushing our custom sealant (PVA glue & water) onto the foam body of the foam body only, avoiding other parts, because SLS-printed Nylon 12 (used for other components) has low absorbency. Any minor absorption in the Nylon 12 could be addressed by sanding its surface to smooth it out.

Main Assembly

To align our 3D printed parts when assembly the car, we designed a custom assembly jig.



11.1: CAD model of jig
Credit: Lunar

Adheres to: CAD accuracy and smooth surfaces goals.

To glue the parts together, we used Loctite Super Glue Control for its strong impact strength and integrity for part assembly, then leaving it to dry over 24 hours.

Putty Filler

To fill gaps within the body and 3D printed parts, we used an Earl's water-based Timbermate putty filler (non-shrinking).

PUTTY FILLER	WEIGHT (g/cm³)	OBSERVATIONS
Deluxe Plastic Putty	1.4	Sticky, slight shrinkage, dried too quick, can only apply thin layers at a time.
Earl's Multi-Filler	1.8	Normal drying time, consistent texture, minimal shrinkage, easy to apply.
Earl's Timbermate Filler	1.2	Smooth, non-shrinking, durable, strong adhesion, consistent, easy to apply.

11.2: Putty Filler Investigation
Credit: Lunar

= Chosen filler



11.3: Newly puttied car
Credit: Lunar

Once applied cautiously and left to dry for 24 hours, any excess putty was sanded away with 280-800 grit sandpaper depending on how much was present. We also had to ensure the sealant was not broken, but if so, it was re applied in damaged areas.

Primer Filler

Primer Filler, Septone Acrylic Spray Putty, was used to fill in the micro gaps and impurities of the assembled surface. 2 rounds of spraying and sanding were done, 2 hours in between, with 4 light layers in each with 15 minutes to dry for each application.

This gave us extremely fine surfaces and only added 1g of weight.



11.4: Newly primer filled car
Credit: Lunar

Painting

Our painting was done with White "Rust-Oleum 2x Ultra Cover Paint+Primer Ultra Matte", which uses special any-angle technology and anti-cramp nozzles, reducing sputtering and risk of damage to smooth surfaces. This paint also acts as a primer to fill in any missed surface impurities.

This was then sanded with 2000 and 3000 grit sandpaper, creating surfaces smoother than we've ever created before.



11.5: The smooth surface achieved
Credit: Lunar

Stickers

We used waterslide laser decals sourced locally to produce the stickers for car. This was used due to their:

- Micro thickness - see figure 11.7.
- Non-impeding presence - they do not interfere with the surface of the car.
- Malleability - can be shaped around all surfaces of the car.
- Low weight - almost 0g.

One problem with this was sharp corners tend to curl and protrude, so we curved all corners prevent this.



11.6: Close-up of waterslide decal stickers
Credit: Lunar

Final Seal

To seal the stickers and apply a final clean finish, we used a Satin Varnish Clear Coat, which also reduced the impact of paint humidity alteration. This Satin finish was chosen over others because:

- Gloss finish - much heavier and damages stickers.
- Matte finish - not aesthetically appealing.

Wheel Manufacturing

For wheel manufacturing we used the Creality H1+ and UW-03 station. When washing we tried methylated spirits but it was very dangerous. So instead we used Isopropyl Alcohol (IPA) resulting in smoother finishes. We washed in 2 baths of IPA, first to remove chunks and then to clean up missed details plus manual agitation. These were then cured using the turn table for 10 minutes

Wheel Assembly

We needed to ensure our wheels were assembled without warping our breakage. This is why we created multiple wheel sets for our interchangeable system involving fully optimised bearings on their performance curve.

To ensure the accuracy of these wheel systems we created another 2 assembly jigs for our double bearings and to fit the wheel onto our axle.



11.7: Axel support jig (left) and double bearing jig (right)
Credit: Lunar

These jigs ensured our stable wheel system and CAD accuracy goals.

Final Weight Calculation

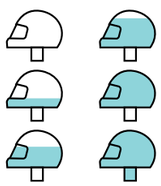
	Foam body	17.47	+	**Paints**	5.3
+	**Sealant**	1.2	+	**Stickers & cleacoat**	0.6
+	**3D printed parts**	16.54	+	**Wheel system**	7.1
+	**Putties**	1	+	**Tether system**	0.8

TOTAL: 50.01g **not including changes during travel*

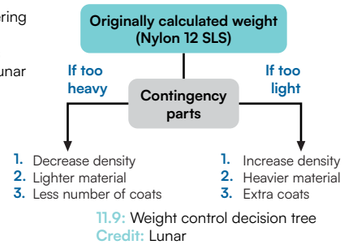
QUALITY ASSURANCE

Final Weight Controls

We developed various contingency parts to allow us to ensure our weight was 50.03-50.08g after possible changes during travel. These were helmets and hubcaps of different materials, coating and porosities, allowing for alterations of 0.01g.



11.8: Differing helment porosities
Credit: Lunar



11.9: Weight control decision tree
Credit: Lunar

Self & 3rd Party Scrutineering

We scrutineered our cars every time we altered the design with past judges, our whole team, Formula 3 engineers, and more. This helped confirm we hadn't missed anything and had a larger assurance in results and tips from industry professionals.

Selection of Cars

We created 6 race-spec cars instead of only 2 in case of possible manufacturing errors that would impact our overarching objectives. Once all completely manufactured, the 2 best quality were our race cars to submit.



11.10: 6 contingency cars
Credit: Lunar

CAD Accuracy (Car Casts)

To confirm that our manufactured car replicated our CAD model as accurately as possible, we created casts of our car at certain sections.



11.11: Car casts
Credit: Lunar

Dust-Free Operations

To prevent dirt and dust interference when painting, we used air purifiers that collect micro particles in the air.

We also used a spray booth when painting, which filters dust away from our working area.



11.12: Air purifier tower
Credit: Lunar

Priming, Paint and Clearcoat Testing

To ensure quality of paints and primers, we initially tested various types and brands on test blocks to ensure we use the highest quality on our final car whilst considering weight.



11.13: Test paint blocks
Credit: Lunar

OUTSOURCING JUSTIFICATION

Make vs Buy - 3D Printing

We outsourced our 3D printing and jigs because they needed to be the highest accuracy, which is best when professionally sourced.

METHOD	MAKE/BUY	COST ($)	LEAD TIME	QUALITY	STRENGTH
SLS Nylon 12	Buy	≈400	6 days	Excellent	Best
FDM ABS	Make	≈30	9 days	Poor	Ok
SLA ABS-Like	Make	≈15	4 hours	Best	Ok

MATERIAL	MAKE/BUY	FRICTION CO-EFFICIENT	STRENGTH
SiC	Buy	0.2	Very high
Alconite	Buy	0.1	Moderate
Resin	Make	0.5	Low

= Chosen material/method

11.19: 3D printing type evaluation
Credit: Lunar

Make vs Buy - Wheels

We considered 3 methods to manufacturing our wheels:

METHOD	WHERE	COST ($)	LEAD TIME	QUALITY
Injection moulding	Outsourcing	≈5000	2 weeks	Best
CNC Turning	Outsourcing	≈1200	2 weeks	Very good
SLA	In-house	15	4 hours	Good

11.9: Wheel Manufacturing investigtion
Credit: Lunar

Despite reduced quality, due to the low cost and high enough structural integrity, it was chosen as our manufacturing method.

= Chosen method

Material: ABS-like Resin.

Bearings

We outsourced our bearings because it was impossible to create our own to industry standard

TYPE	MAKE/BUY	COST ($)	LEAD TIME	QUALITY
Full Ceramic	Buy	20-40	5-9 days	ABEC 7-9
Hybrid-Steel	Buy	20-80	9-10 days	ABEC 5-9
Full Steel	Buy	5-15	1-3 days	ABEC 3-5
Resin	Make	10	5-6 days	Not Rated

11.10: Justification of bearings
Credit: Lunar

The Full ceramic bearings were used because of their long duration, good strength, heat dissipation, and ABEC 7 rating.

WORKPLACE SAFETY

DANGER	STAGE	RISK	STANDARD OPERATING PROCEDURES (SOPs)
Inhaling harmful particles fumes	Inhaling harmful particles fumes	High	The air was purified from any harmful micro particles when Milling, SLA printing, or painting. A mask was also used as a precaution. Spraying booth was also utilised to recycle harmful air into safe air.
Skin contact with harmful materials	SLA Printing, Paint particles	Medium	Used Latex gloves to block direct contact with skin and stayed away from direct UV exposure around SLA printing.
Hearing Damage	Milling	Low	Used sound industry earmuffs when spending extended periods of time around the machine.
Cuts & lacerations	Hand manufacturing	Medium	It was ensured that where possible cutting was away from the person and the workpiece was secured and not loose.
Entanglement or Impact	Milling, Bearing burnishing,	Low	Work aprons were used to restrict free clothing like ties. Milling was also supervised by experienced teachers with qualifications. Pre-machining check-list was ticked off before milling.
Eye irritation	Painting, SLA printing	Low	Goggles were used to block vapours and particles from contact with the eyes causing irritation.

11.17: Workplace safety considerations
Credit: Lunar